Artificial Intelligence
Introduction and examples

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BLIND SEARCH ALGORITHMS

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Depth-first search

1. QUEUE ←path only containing the root;

2. WHILE (QUEUE is not empty AND goal is not reached)
   - Remove the first path from the QUEUE
   - Create new paths (to all children);
   - Reject the new paths with loops;
   - Add the new paths to the **front** of QUEUE

3. IF (goal reached)
   - THEN SUCCESS
   - ELSE failure

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Breadth-first search

1. QUEUE ←path only containing the root;

2. WHILE (QUEUE is not empty AND goal is not reached)
   - Remove the first path from the QUEUE;
   - Create new paths (to all children);
   - Reject the new paths with loops;
   - Add the new paths to the **back** of QUEUE

3. IF (goal reached)
   - THEN SUCCESS
   - ELSE failure

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Non-deterministic search

1. QUEUE ←path only containing the root;

2. WHILE (QUEUE is not empty AND goal is not reached)
   - Remove the first path from the QUEUE
   - Create new paths (to all children);
   - Reject the new paths with loops;
   - Add the new paths in random places in QUEUE

3. IF (goal reached)
   - THEN SUCCESS
   - ELSE failure

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Iterative deepening search: depth-limited search

1. QUEUE ←path only containing the root;
   DEPTH ← number;

2. WHILE (QUEUE is not empty AND goal is not reached)
   - Remove the first path from the QUEUE
   - IF path has length smaller than DEPTH:
     - Create new paths (to all children);
     - Reject the new paths with loops;
     - Add the new paths to the **front** of QUEUE

3. IF (goal reached)
   - THEN SUCCESS
   - ELSE failure

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Iterative deepening search

1. DEPTH = 1

2. WHILE goal is not reached
   - perform Depth-limited search
   - DEPTH = DEPTH + 1

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Finally: what blind search algorithm to choose?

Criterium	DFS	BFS	IDS
Complete?	Yes, if (1) no infinite branches (2) no cycles	Yes, always	Yes, always
Shortest path?	No, maybe by accident	Yes, always	Yes, always
Speed?	$O(b^d)$	$O(b^m)$	$O(b^m)$
Memory?	$O(bd)$	$O(b^m)$	$O(bm)$

Iterative Deepening Search!

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HEURISTIC SEARCH ALGORITHMS

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Hill climbing

1. QUEUE ←path only containing the root;
   heuristic h ← function;

2. WHILE (QUEUE is not empty AND goal is not reached)
   - Remove the first path from the QUEUE
   - Create new paths (to all children);
   - Reject the new paths with loops;
   - Sort the new paths using the heuristic h
   - Add the new paths to the **front** of QUEUE

3. IF (goal reached)
   - THEN SUCCESS
   - ELSE failure

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Beam search

1. QUEUE ←path only containing the root;
   heuristic h ← function;
   WIDTH ← number;

2. WHILE (QUEUE is not empty AND goal is not reached)
   - Remove **all paths** from the QUEUE
   - Create new paths (to all children);
   - Reject the new paths with loops;
   - Sort the new paths using the heuristic h
   - (Remove the paths with no successors)
   - Add **the WIDTH best paths** to QUEUE

3. IF (goal reached)
   - THEN SUCCESS
   - ELSE failure

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Greedy search

1. QUEUE ←path only containing the root;
   heuristic h ← function;

2. WHILE (QUEUE is not empty AND goal is not reached)
   - Remove the first path from the QUEUE
   - Create new paths (to all children);
   - Reject the new paths with loops;
   - Add the new paths and sort the entire QUEUE using heuristic h

3. IF (goal reached)
   - THEN SUCCESS
   - ELSE failure

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OPTIMAL SEARCH ALGORITHMS

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Uniform cost algorithm

1. QUEUE ←path only containing the root;

2. WHILE (QUEUE is not empty AND goal is not reached)
   - Remove the first path from the QUEUE
   - Create new paths (to all children);
   - Reject the new paths with loops;
   - Add the new paths to the QUEUE and sort the entire QUEUE (by accumulated cost)

3. IF (goal reached)
   - THEN SUCCESS
   - ELSE failure

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Optimal uniform cost algorithm

1. QUEUE ←path only containing the root;

2. WHILE (QUEUE is not empty AND first path does not reach goal)
   - Remove the first path from the QUEUE
   - Create new paths (to all children);
   - Reject the new paths with loops;
   - Add the new paths to the QUEUE and sort the entire QUEUE (by accumulated cost)

3. IF (goal reached)
   - THEN SUCCESS
   - ELSE failure

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Branch-and-bound extended uniform cost algorithm

1. QUEUE ←path only containing the root;
   BOUND ← very large number;

2. WHILE (QUEUE is not empty AND **first path does not reach goal**)
   - Remove the first path from the QUEUE
   - Create new paths (to all children);
   - Reject the new paths with loops;
   - Add the new paths to the QUEUE and sort the entire QUEUE (by accumulated cost)
   - BOUND ← lowest cost of new paths that reach goal and have cost lower than BOUND;
   - Remove paths from QUEUE with cost larger than BOUND

3. IF (goal reached)
   - THEN SUCCESS
   - ELSE failure

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Estimate-extended uniform cost algorithm

1. QUEUE ←path only containing the root;
   heuristic h ← function;

2. WHILE (QUEUE is not empty AND **first path does not reach goal**)
   - Remove the first path from the QUEUE
   - Create new paths (to all children);
   - Reject the new paths with loops;
   - Add the new paths to the QUEUE and sort the entire QUEUE (by f = cost + h)

3. IF (goal reached)
   - THEN SUCCESS
   - ELSE failure

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A* algorithm

1. QUEUE ← path only containing the root;
   heuristic h ← function;

2. WHILE (QUEUE is not empty AND first path does not reach goal)
   - Remove the first path from the QUEUE
   - Create new paths (to all children);
   - Reject the new paths with loops;
   - Add the new paths to the QUEUE and sort the entire QUEUE (by f = cost + h)
   - If QUEUE contains:
     - path P terminating in I, with cost_P
     - path Q containing I, with cost_Q

     and if cost_P ≥ cost_Q THEN delete path P

3. IF (goal reached)
   - THEN SUCCESS
   - ELSE failure

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Questions

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